

**OPTIMIZING INVENTORY
MANAGEMENT FOR RETAIL
BUSINESS USING THE
SUPERSTORE**

1. INTRODUCTION

Superstore Sales Dashboard also helps automate sales forecasting and inventory management through the availability of real-time data on geographic sales, sales performance, and units of products sold. It is a valuable tool that helps in decision-making and working procedure automation. It facilitates easy monitoring and tracking of matters like profitability, product demand, and sales trends, and it is simple to project future requirements and mobilize resources. The stakeholders are sales managers, inventory managers, and executives who can utilize this data to make informed decisions, maximize sales efforts, and raise regional stock levels. Through the presentation of key business insights graphically, the dashboard enables swift, practical action in dynamic retail environments.

2. DATA PRODUCT AND INSIGHTS

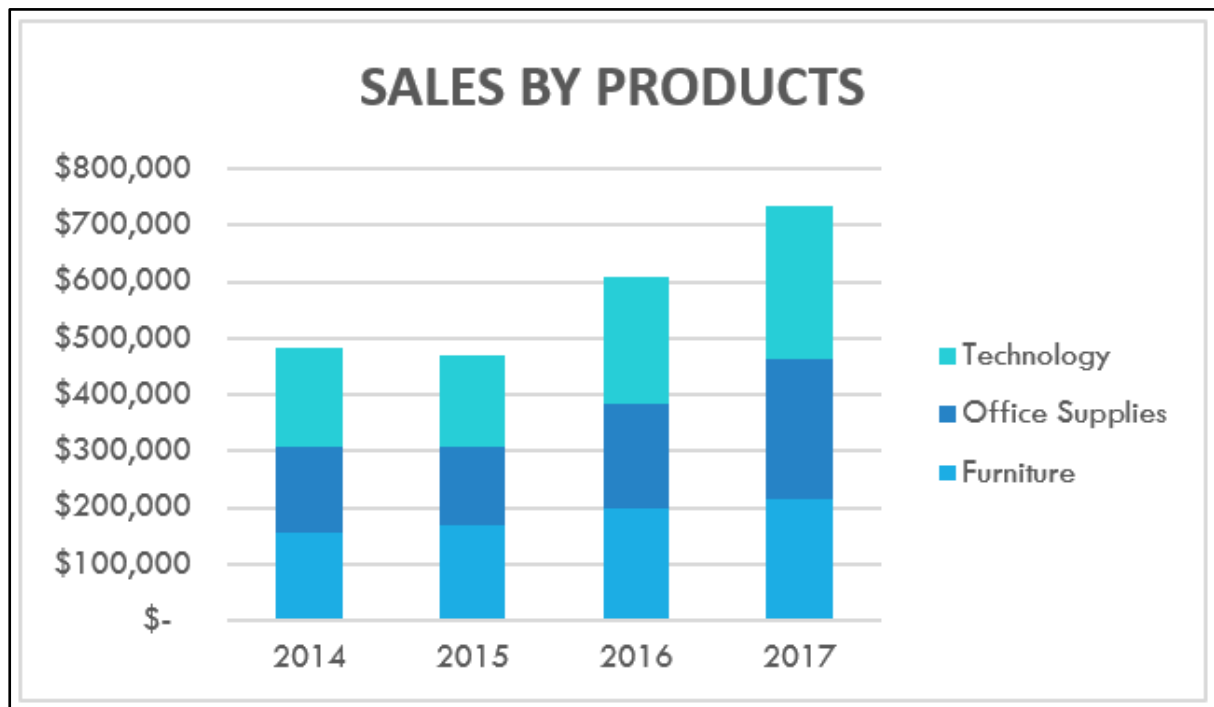
The Superstore Sales Dashboard integrates multiple visualizations designed to provide actionable insights into sales performance and product demand. These insights help stakeholders monitor key metrics and optimize retail operations :

Figure 1 - Superstore Dashboard



2.1 SALES BY PRODUCTS

Figure 2 - Sales by Products

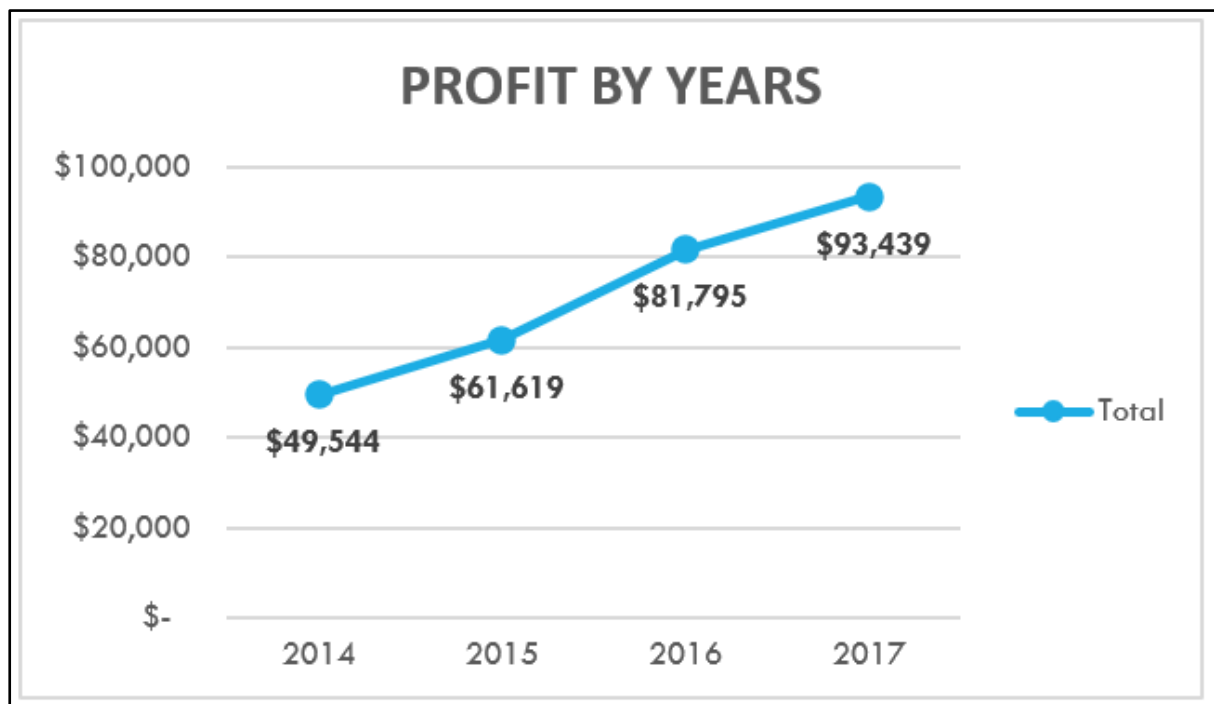


The chart breaks down total sales by product category (Technology, Office Supplies, and Furniture) between 2014 and 2017. It allows managers to see how each category is trending over time and to ascertain whether the sales strategy is sufficiently project performance in each category.

2.2 PROFIT BY YEARS

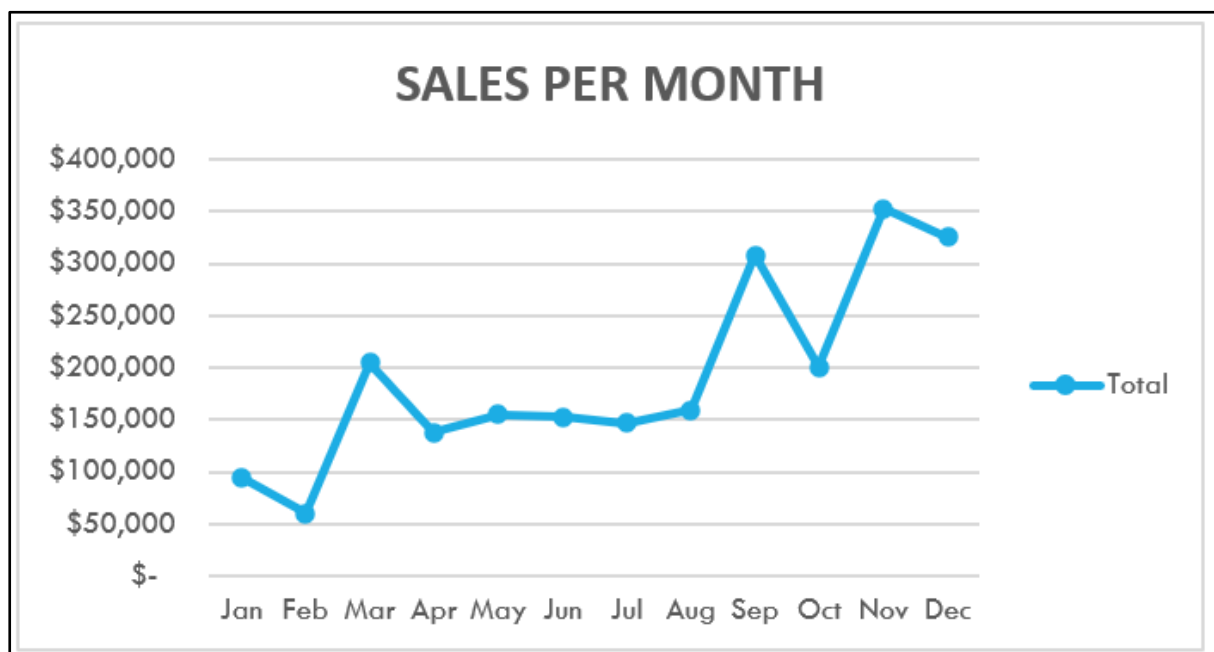
The line chart demonstrates a year-on-year profit increase, from \$49,544 in 2014 to \$93,439 in 2017. From the profitability trends, managers can evaluate the business's financial health and identify the years when profits have risen or dropped considerably to make informed decisions.

Figure 3 - Profit by Years



2.3 SALES PER MONTH

Figure 4 - Sales per Month

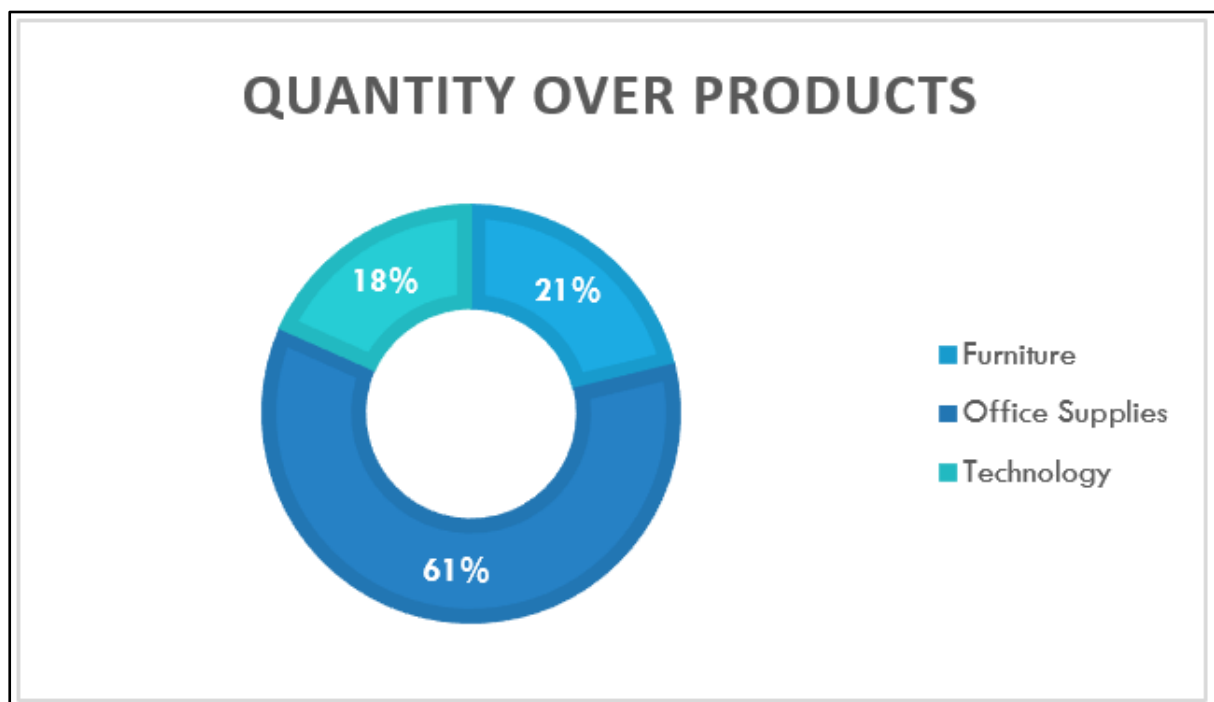


The sales per month graph presents monthly sales trends so that users can identify seasonal peaks. If sales, for instance, reach the highest level in November and December, businesses can schedule holiday season inventory needs more accurately.

2.4 QUANTITY OVER PRODUCT

The pie chart shows how the total quantity sold is distributed across each product category. As indicated herein, Technology has 61%, and this discovery helps managers know where the demand for products is to stock to prevent overstocking in less demanded product categories like Furniture.

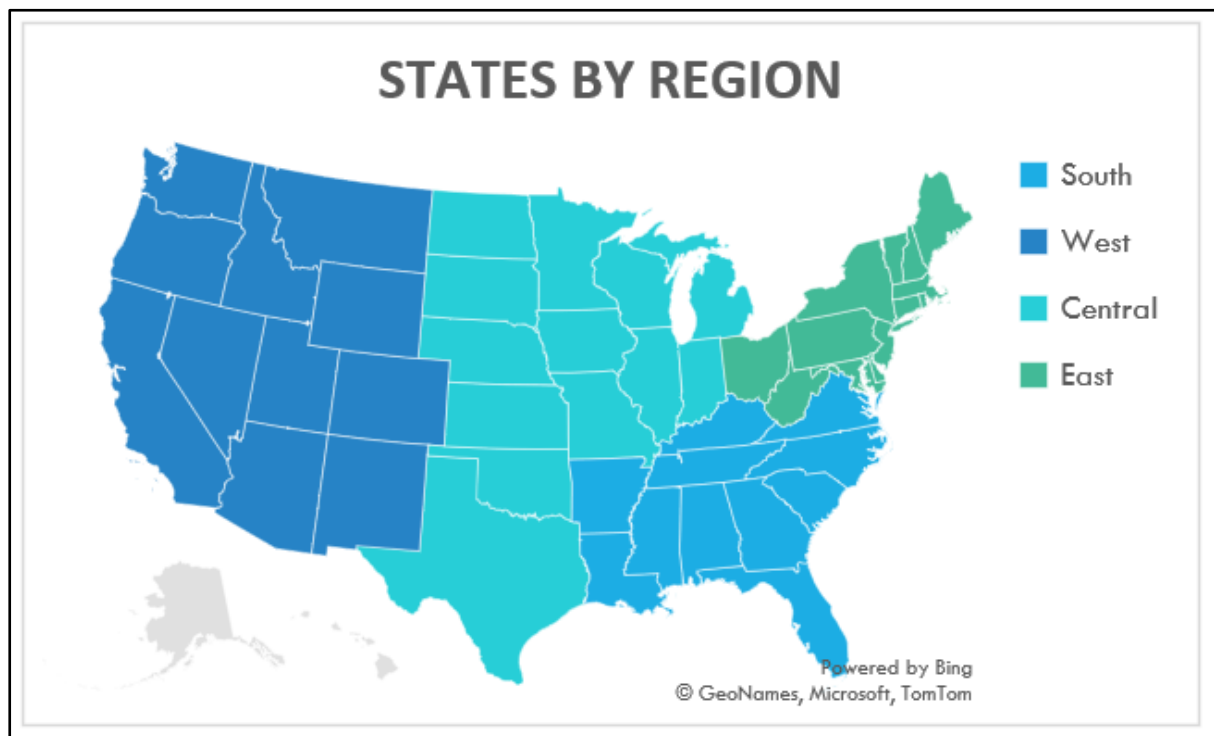
Figure 5 - Quantity over Products



2.5 STATES BY REGION

This map shows sales performance by four regions (South, West, Central, East), which enables the identification of strong and weak-performing regions. Regional data can be used in resource allocation, planning regional promotions, and focusing marketing efforts where needed.

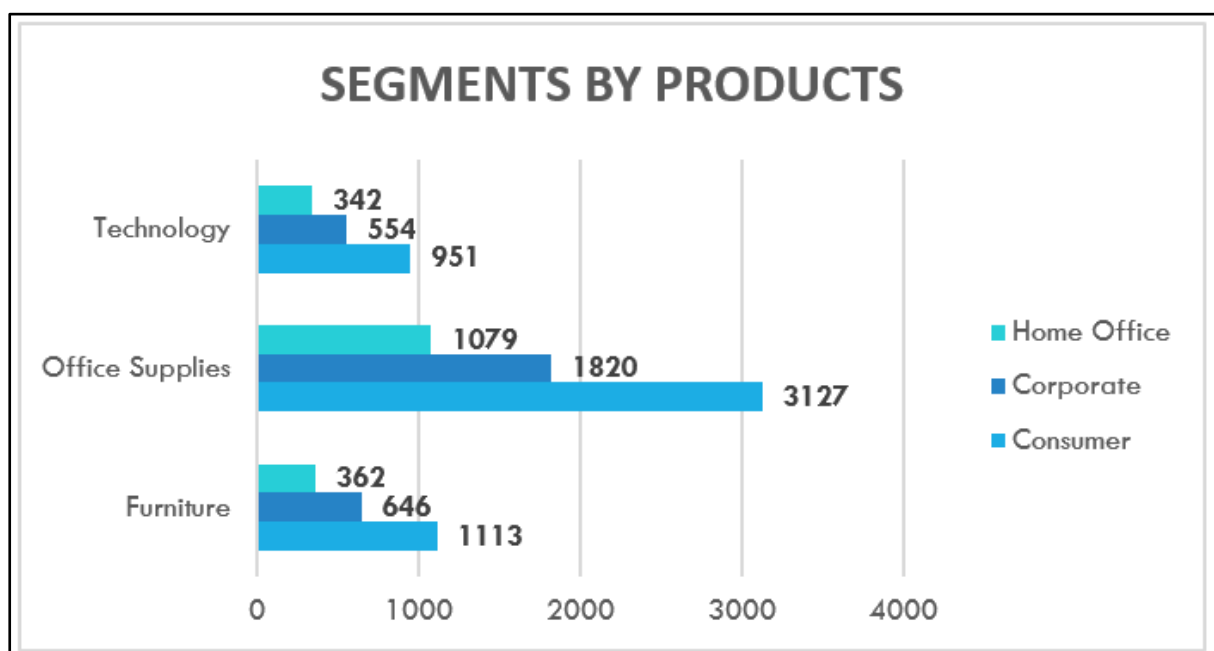
Figure 6 - States by Region



2.6 SEGMENTS BY-PRODUCTS

This chart shows how each product category is sold via customer segments (Consumer, Corporate, Home Office). The highest sales segment is the Home Office, which can guide targeted marketing campaigns.

Figure 7 - Segments by Products



3. TARGET USERS

- ❑ **Sales Managers:** This group benefits from sales trends and category performance data. Sales managers can use the Sales by Products and Sales Per Month charts to assess how products perform across various periods and adjust strategies for high-performing categories like Technology. This information allows them to plan promotions and adjust strategies for underperforming categories.
- ❑ **Inventory Manager:** Sales forecasts and quantity sold by category are crucial for inventory managers. Using the Quantity Over Products chart, they can ensure adequate stock for Technology and Office Supplies while minimizing overstocking in low-demand categories like Furniture. The Sales Per Month chart also helps them prepare for peak sales months by adjusting stock levels in advance.
- ❑ **Executives:** Executives benefit from the top-line data provided by Profit by Year and Regional Sales Performance. It allows executives to track the combined performance, profit trends, and regional victory to make expansion, resource allocation, or thrust marketing decisions for prosperous regions. Strategic choices and long-term strategic planning are made based on such information.

4. FUTURE IMPROVEMENTS

Currently, the dashboard is associated with past sales history and, while helpful, is not in real-time. If real-time data from the sales system was utilized, the dashboard would be interactive, and users could respond in real-time to shifting demand and sales performance. This would be particularly valuable to inventory managers who must make decisions based on real-time data. While the dashboard does have fundamental sales trends, more sophisticated machine learning algorithms or even better time series forecasts can forecast future sales more precisely.

5. CONCLUSION

Superstore Sales Dashboard is a robust graphical tool that provides real-time data on sales performance, product demand, and geographic sales patterns. Interactive charts and filters help inventory managers, sales managers, and executives make informed decisions, control inventory levels, and optimize sales strategy. The key functionalities of the dashboard, such as category sales, tracking of profit, and performance in regions, reflect the users' requirements and increase business effectiveness. Future additions, such as real-time data integration and sophisticated forecasting, will further augment the value of the dashboard and enable even better decision-making for the retail business.